

YAML Tutorial

①

What is YAML? [YAML Ain't Markup Language]

Despite the name it is not a markup language like HTML.

It is a human readable way to store configuration of data.

Ex

```
database_host = "localhost"
database_port = 5432
database_name = "mydb"
```

} Python code.

database :

```
  host: localhost
  port: 5432
  name: mydb
```

} YAML.

Why YAML?

Let suppose you are building ML project

without YAML

↳

learning_rate = 0.01

epochs = 50

batchsize = 32

dataset = "spam.csv"

python

Next month you want

learning_rate = 0.01

epochs = 100

So, you have to edit your python code

instead store everything in YAML.

So, to change we don't need to open our code.

How

person = { "Name": "Himanshu",

"Age": 28

"City": Bangalore }

python code

in YAML

we don't have { } bracket

every thing is manage by indentation

Person:

name: Himanshu

Age: 28

City: BLP

YAML

②

every thing under 2 space.

Cases (use case)

① Key-value pair.

name: John Doe

age: 30

YAML primarily use key-value pairs, where the key and value are separated by colon and a space.

② Hierarchy through indentation:-

- YAML uses indentation (space or tab) to represent nested structure like dictionaries in python.

address:

Street: 123 main st

City: Anytown

③ Lists:-

④

List in YAML are denoted by a hyphen followed by a space. Each item in the list is placed on a new line.

fruits:

- Apple
- Orange
- Banana

Indentation + '-'

new line for each element.

④ Comments:-

Comments in YAML start with a # and can be placed anywhere in the document.

This is a comment
name: John Doe.

⑤ Data type:-

YAML supports various data types including strings, numbers, boolean, null and complex type like lists and dictionaries.

⑤ * String does not need to be quoted unless they contain special characters.

Ex

age: 30

married: true

children: null

Name: John Doe

email: "123@gmail.com"

⑥ multiline string :-

YAML allows for multiline strings using
| (preserve newline) or > (fold newline into space)

Ex

description: |

This is a
multiline string

⑦ ~~Dictionary~~ Anchors & Aliases:-

~~YAML~~ ~~allows~~ ~~for~~ ~~reusable~~ ~~strings~~

Anchors (&) and aliases (*) allow you to reuse content. An anchor defines a value and an alias refers back to it.

Example

default: & default settings → creating alias
timeout: 30 (Anchoring)
retries: 3

Server:

<<: * default settings → re-using it.
host: Server1.example.com

⑧ YAML Vs json:

YAML is a superset of json

means → any valid json is a valid
YAML.

YAML is more user friendly due to
its readability and concise syntax.

⑨ File extensions and usages

- YAML file typically .yaml or .yml
- used for configuration files.
- Data exchange b/w languages.
- more in Devops, CI/CD pipeline.